

ODS Formula

At a local optimum x^* ,

$$\nabla f(x^*) = 0$$

This means: the gradient (slope) must be zero $\Rightarrow$ stationary point

♦ Second Order Necessary Condition (SONC):

If x^* is a local minimizer, and $\nabla f(x^*) = 0$, then:

$$\nabla^2 f(x^*) \succeq 0 \quad (\text{Hessian is positive semi-definite})$$

♦ Second Order Sufficient Condition (SOSC):

If $\nabla f(x^*) = 0$ and

$$\nabla^2 f(x^*) \succ 0 \quad (\text{Hessian is positive definite})$$

then x^* is a strict local minimizer.

🧠 Interpretation:

Condition	What it means
$\nabla f = 0$	Flat slope (stationary point)
Hessian $\succeq 0$	Local minimum
Hessian $\preceq 0$	Local maximum
Hessian indefinite	Saddle point

ODS Formula

✓ Newton's Method — Single Variable

- ♦ Goal: **Find a root of $f'(x) = 0$, i.e., stationary point (min/max)**
- ♦ Iteration Formula:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

- ♦ Conditions:
 - $f \in C^2$ (twice continuously differentiable)
 - Converges quadratically near the solution if $f''(x^*) \neq 0$
 - Works best if x_0 is close to the minimizer
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✓ Newton's Method — Multivariable

- ♦ Goal: **Solve $\nabla f(x) = 0$**
- ♦ Iteration Formula:

$$x_{k+1} = x_k - H^{-1}(x_k) \cdot \nabla f(x_k)$$

Where:

- $\nabla f(x_k)$: gradient vector
 - $H(x_k) = \nabla^2 f(x_k)$: Hessian matrix
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ODS Formula

✓ Steepest Descent Method

- ◆ Goal: **Minimize $f(x)$ iteratively using gradient information**
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- ◆ Update Rule:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

Where:

- $\nabla f(x_k)$: gradient (points uphill)
 - $-\nabla f(x_k)$: steepest descent direction
 - α_k : step size (can be fixed or from exact/line search)
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- ◆ Stopping Criteria:

$$\|\nabla f(x_k)\| < \epsilon$$

✓ Directional Derivatives

- ◆ Definition:

Rate of change of $f(x)$ at point x in direction d

- ◆ Formula:

$$D_d f(x) = \nabla f(x)^T d$$

Where:

- d : unit direction vector (i.e., $\|d\| = 1$)
 - $\nabla f(x)$: gradient vector
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- ◆ Steepest Descent Direction:

Direction $d = -\frac{\nabla f(x)}{\|\nabla f(x)\|}$

This is the direction in which f decreases most rapidly.

ODS Formula

✔ Quasi-Newton Method

- ◆ Goal: **Minimize $f(x)$ using gradient + Hessian approximation**

- ◆ Update Rule:

$$x_{k+1} = x_k - B_k^{-1} \nabla f(x_k)$$

Where:

- B_k is an approximation to the Hessian or its inverse
- Typically initialized with $B_0 = I$

📘 Common Quasi-Newton Variants:

- ◆ DFP (Davidon-Fletcher-Powell) Update:

$$B_{k+1} = B_k + \frac{s_k s_k^T}{s_k^T y_k} - \frac{B_k y_k y_k^T B_k}{y_k^T B_k y_k}$$

Where:

- $s_k = x_{k+1} - x_k$
- $y_k = \nabla f_{k+1} - \nabla f_k$

- ◆ Inverse BFGS (used in practice):

If we store the inverse Hessian approximation $H_k = B_k^{-1}$, then:

$$H_{k+1} = \left(I - \frac{s_k y_k^T}{y_k^T s_k} \right) H_k \left(I - \frac{y_k s_k^T}{y_k^T s_k} \right) + \frac{s_k s_k^T}{y_k^T s_k}$$

ODS Formula

✓ Conjugate Gradient Method (CGM)

- ◆ Used to minimize:

$$f(x) = \frac{1}{2} x^T A x - b^T x$$

Where:

- A is symmetric positive definite (SPD)
- $x, b \in \mathbb{R}^n$

- ◆ Initialization:

$$\begin{aligned} x_0 &= \text{initial guess} \\ r_0 &= b - Ax_0 \quad (\text{residual}) \\ d_0 &= r_0 \quad (\text{initial search direction}) \end{aligned}$$

- ◆ Iterative Steps:

For $k = 0, 1, 2, \dots$

1. Step size:

$$\alpha_k = \frac{r_k^T r_k}{d_k^T A d_k}$$

2. Update iterate:

$$x_{k+1} = x_k + \alpha_k d_k$$

3. Update residual:

$$r_{k+1} = r_k - \alpha_k A d_k$$

4. Direction update coefficient:

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}$$

5. Update direction:

$$d_{k+1} = r_{k+1} + \beta_k d_k$$

◆ Stopping Condition:

$$\|r_k\| < \epsilon \quad (\text{gradient close to 0})$$

✔ Conjugate Directions

◆ Definition:

Vectors $d_1, d_2, \dots, d_n$ are A-conjugate (or A-orthogonal) if:

$$d_i^T A d_j = 0 \quad \text{for } i \neq j$$

This means:

- Each direction is orthogonal with respect to A
- Conjugate directions generalize orthogonality in non-Euclidean inner products

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ODS Formula

Lagrange Multipliers

Given:

$$\min f(x), \quad \text{subject to } g_i(x) = 0, \quad i = 1, \dots, m$$

Lagrangian Function:

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

Stationarity Condition:

$$\nabla_x \mathcal{L}(x, \lambda) = \nabla f(x) + \sum_{i=1}^m \lambda_i \nabla g_i(x) = 0$$

Primal Feasibility:

$$g_i(x) = 0, \quad i = 1, \dots, m$$

Karush-Kuhn-Tucker (KKT) Conditions

Given:

$$\min f(x), \quad \text{subject to } g_i(x) \leq 0, \quad i = 1, \dots, m; \quad h_j(x) = 0, \quad j = 1, \dots, p$$

Lagrangian Function:

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x)$$

KKT Conditions:

1. Stationarity:

$$\nabla f(x) + \sum_{i=1}^m \lambda_i \nabla g_i(x) + \sum_{j=1}^p \mu_j \nabla h_j(x) = 0$$

2. Primal Feasibility:

$$g_i(x) \leq 0, \quad h_j(x) = 0$$

3. Dual Feasibility:

$$\lambda_i \geq 0$$

4. Complementary Slackness:

$$\lambda_i g_i(x) = 0$$

ODS Formula

✓ Simplex

- ◆ All constraints ($\leq$)

Minimizing:

- Optimal: $C_j - Z_j \geq 0$
- Pivot:
 - Column $\rightarrow$ Most negative $C_j - Z_j$
 - Row $\rightarrow$ Least positive ratio (ignore ≤ 0)

Maximizing:

- Optimal: $C_j - Z_j \leq 0$
 - Pivot:
 - Column $\rightarrow$ Largest positive $C_j - Z_j$
 - Row $\rightarrow$ Least positive ratio (ignore ≤ 0)
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✓ Big M Method

- ◆ When constraints include ($\geq$), ($=$)
 - $\leq$: Add Slack
 - $\geq$: -Slack + Artificial
 - $=$: Add Artificial

Minimizing:

- $Z = \text{Original} + M \times \text{Artificials}$ (Add +M terms)

Maximizing:

- $Z = \text{Original} - M \times \text{Artificials}$ (Use -M terms)

➡ Rest is same as simplex

ODS Formula



Corrected Barrier Method

Objective:

$$\min f(x), \quad \text{subject to } g_i(x) < 0$$

Barrier Function (Log Barrier):

$$B(x,r) = f(x) - \frac{1}{r} \sum_{i=1}^m \ln(-g_i(x))$$

Condition:

$$r \rightarrow 0^+$$

Penalty Method

Given:

$$\min f(x), \quad \text{subject to } g_i(x) \leq 0, \ h_j(x) = 0$$

Penalty Function:

$$P(x,r) = f(x) + r \left(\sum_{i=1}^m \max(0, g_i(x))^2 + \sum_{j=1}^p h_j(x)^2 \right)$$

Unconstrained Reformulation:

$$\min P(x,r)$$

Condition:

$$r \rightarrow \infty$$

Barrier Method

Given:

$$\min f(x), \quad \text{subject to } g_i(x) < 0$$

Barrier Function (Log Barrier):

$$B(x,r) = f(x) - \frac{1}{r} \sum_{i=1}^m \ln(-g_i(x))$$

Unconstrained Reformulation:

$$\min B(x,r)$$

Condition:

$$r \rightarrow \infty$$

ODS Formula

Frank-Wolfe Algorithm

Given:

$$\min_{x \in \mathcal{D}} f(x), \quad \text{where } f \text{ is convex and differentiable, and } \mathcal{D} \text{ is convex}$$

Step-by-step (pure formula style)

1. Initialization:

$$x_0 \in \mathcal{D}, \quad k = 0$$

2. Linear Subproblem:

$$s_k = \arg \min_{s \in \mathcal{D}} \nabla f(x_k)^T s$$

3. Direction:

$$d_k = s_k - x_k$$

4. Step Size (line search or fixed):

$$\gamma_k = \arg \min_{\gamma \in [0,1]} f(x_k + \gamma d_k)$$

5. Update:

$$x_{k+1} = x_k + \gamma_k d_k$$

6. Repeat until convergence